POINTS TO REMEMBER

- F-statistic is the ratio of two independent sample variances
- If X and Y are two independent χ^2 variates with m and n degrees of freedom respectively, then $F = \frac{X/m}{Y/n}$ is said to follow F distribution with (m, n) degrees of freedom.
- Two independent random samples of size m and n are taken from Normal populations. Then the statistic $F = \frac{s_X^2}{s_Y^2}$ is a random variable following the F-distribution with $m-1$ and $n-1$ degrees of freedom.
- According to R.A. Fisher, ANOVA is the "Separation of variance, ascribable to one group of causes from the variance ascribable to other groups".
- One-way ANOVA is used to compare means in more than two groups.
- Two-way ANOVA is used to compare means in more than two groups, controlling another variable.
- Assumptions required for ANOVA are:
  - The observations follow normal distribution.
  - Experimental units assigned to treatments are random.
  - The sample observations are independent.
  - The population variances of the groups are unknown but are assumed to be equal.

EXERCISE 3

CE136V

I. Choose the best Answer.

1. ANOVA was developed by

   (a) S.D. Poisson (b) Karl Pearson

   (c) R.A. Fisher (d) W.S. Gosset

2. ANOVA technique originated in the field of

   (a) Industry (b) Agriculture

   (c) Medicine (d) Genetics

3. One of the assumptions of ANOVA is that the population from which the samples are drawn is

   (a) Binomial (b) Poisson (c) Chi-square (d) Normal

4. In one-way classification the total variation can be split into

   (a) Two components (b) Three components

   (c) Four components (d) Only one components

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